Assignment Sheet 3

Problem 1
In this assignment you are to improve the query times of the CH you have constructed in the last assignment by computing so-called *distance closures* (see original paper in the Ilias course).

You can use your own CH, but we will also provide you with a precomputed CH just in case. Two precomputed CHs (Regierungsbezirk Stuttgart and Germany) can be found at `https://fmi.uni-stuttgart.de/alg/research/stuff/` and `https://fmi.uni-stuttgart.de/files/alg/data/graphs/germany.ch.bz2`.

The preprocessing for the given CH of the Germany graph must not take more than 30min on the hardware specified for the previous assignments, query times must not exceed 100μs per query for $k \approx 4$ with moderate space consumption – note that pruning of the additional closure edges is mandatory.

Hint: if the distance checks are too expensive using CH queries only, you might be able to speed up the construction considerably by precomputing hub labels which is feasible for the graphs considered here even though it defies the purpose of distances closures :). Alternatively you might use bootstrapping for the distance checks of the closure edges – similar to HL bootstrapping.

As usual submit your code as specified on the last assignment sheet.